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M.Tech.(PT) DEGREE EXAMINATION, JUNE 2025

Second And Fourth Semester

20PITE53J - TIME SERIES FORECASTING

(For the candidates admitted during the academic year 2020-2021 to 2023-2024)

Time: 3 hours

Max. Marks: 100

PART - A (10 × 2 = 20 Marks)

Answer all Questions

Marks BL CO

- | | | | |
|--|---|---|---|
| 1. What is univariate and bivariate time series? | 2 | 1 | 1 |
| 2. Define forecasting. | 2 | 1 | 1 |
| 3. Write double exponential smoothing formula. | 2 | 1 | 2 |
| 4. How many parameters used in triple exponential smoothing? What are its range? | 2 | 1 | 2 |
| 5. What are the most common method to check stationarity? | 2 | 1 | 3 |
| 6. What is the use of PACF? | 2 | 1 | 3 |
| 7. What is stationarity and causality? | 2 | 1 | 4 |
| 8. What do you mean by order of AR and MA model? | 2 | 1 | 4 |
| 9. Write a structure of VARMA(2,1) model. | 2 | 1 | 5 |
| 10. Write the structure of VAR(2) model. | 2 | 1 | 5 |

PART - B (5 × 16 = 80 Marks)

Answer all Questions

Marks BL CO

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|--|----|---|---|
| 11. (a) Fit the trend by straight line by the method of least squares to the following data. Also estimate y at x=1995.
Year x : 1982 1983 1984 1985 1986 1987 1988 1989
No. of fluctuations f : 23 26 28 32 20 12 12 10
(OR)
(b) (i) Write detailed notes on qualitative and quantitative methods of forecasting.
(ii) Calculate the i-yearly (i = 3, 5, 7) moving averages for the following data.
Year x : 1991 1992 1993 1994 1995 1996 1997
Sales f : 102 105 114 110 108 116 112 | 16 | 1 | 1 |
| 12. (a) Find the Double Exponential Smoothing Forecast values for the following data by taking $\alpha = 0.1, 0.5$ and $\beta = 0.2, 0.3$
Year : 2001 2002 2003 2004 2005 2006 2007 2008 2009 2010 2011 2012
2013 2014 2015 2016
xt : 33.48 39.02 41.39 41.60 44.66 46.95 48.73 51.49 50.03 60.64 63.36
66.36 68.20 68.12 69.78 72.60
(OR)
(b) Find the Single Exponential Smoothing Forecast values for the following data by taking $\alpha = 0.1, 0.2, 0.3, 0.5$
Year: 2001 2002 2003 2004 2005 2006 2007 2008 2009 2010 2011 2012
2013 2014 2015 2016
xt : 33.48 39.02 41.39 41.60 44.66 46.95 48.73 51.49 50.03 60.64 63.36
66.36 68.20 68.12 69.78 72.60 | 16 | 3 | 2 |
| 13. (a) Perform ADF Test by describing the data as an AR(2) model with OLS estimation
t : 1 2 3 4 5 6 7 8 9 10 11 12
xt : 71 70 69 68 64 65 72 78 75 75 75 70 | | | |

(OR)

16 3 3

(b) Compute the correlation, ACF, ACVF, PACF values for the following data.

Also plot the same:

t : 1 2 3 4 5 6 7 8 9 10 11 12

xt : 71 70 69 68 64 65 72 78 75 75 75 70

14. (a) (i) Discuss the Stationarity, Causality, Invertibility of the model

16 3 4

$$x_t = 0.1x_{t-1} + 0.3x_{t-2} + \epsilon_t + \epsilon_{t-1} + 0.25\epsilon_{t-2}$$

(ii) Describe special features of ARIMA model with examples.

(OR)

(b) (i) Compute/Calculate AIC value for the data

t : 0 1 2 3 4 5 6

y : 23 54 65 78 45 67 97

by assuming ARMA(1,1) model with parameters

$$\phi_1 = 1.5, \theta_1 = 0.6, \sigma^2 = 3.2.$$

(ii) Describe the structure of SARIMA model

15. (a) Consider the VARMA(2,1) model $z_t = \phi_1 z_{t-1} + \phi_2 z_{t-2} + \epsilon_t - \theta_1 \epsilon_{t-1}$,

16 3 5

$$\phi_1 = \begin{bmatrix} -0.8 & 0.1 \\ 1 & -0.6 \end{bmatrix}, \quad \phi_2 = \begin{bmatrix} -0.6 & 0.7 \\ 0 & -0.2 \end{bmatrix}$$

$$\theta_1 = \begin{bmatrix} -0.8 & 0 \\ 0 & -1.2 \end{bmatrix}, \quad \Sigma_\epsilon = \begin{bmatrix} 4 & 2 \\ 2 & 5 \end{bmatrix}$$

Calculate/Compute of $\gamma_0, \gamma_1, \gamma_2$, and ρ_1, ρ_2 , and ρ_3

(OR)

(b) (i) Write a detailed description about VARMA model.

(ii) Describe VAR usage by flow chart.

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